



03 ★ Tool Design ★



NEXUS Travel Rule Control Tower

A jurisdiction-aware RegTech prototype for Coinbase-style digital payment token transfers

The same USD 1,500 transfer can be operationally low-friction in the US and evidence-heavy in Singapore. The product decision is political: optimize speed, or enforce the strictest applicable rule.

USD 3,000

US Travel Rule threshold used in the prototype

S\$1,500

Singapore enhanced information threshold

287

synthetic transfers enhanced in SG but not US



Why this product is right for NEXUS?

Commercially useful because fragmentation creates recurring compliance work, not a one-off advisory memo.

NEXUS sells infrastructure to CCOs at exchanges such as Coinbase. The honest customer promise is not financial inclusion; it is lower regulatory surprise and faster evidence production when rules diverge.

PRODUCT FIT

Compliance policy

Engineering controls

Audit evidence

WHY A SPREADSHEET FAILS?

A spreadsheet can compare thresholds, but it cannot reliably version rules, explain data-quality blocks, preserve an audit trail, or route unhosted-wallet exceptions to accountable human review.

The product sits between written policy and transaction execution. It turns legal thresholds into versioned machine-readable routing decisions.

NEXUS wins when fragmentation persists. That is a conflict of interest, so the design must be frank about boundaries, bias and human escalation.



The regulatory divergence is a working parameter

The rule engine stores thresholds, required fields and wallet treatment as jurisdiction data.

THRESHOLD LADDER



US CONFIGURATION

- Below USD 3,000: monitor unless independent AML risk is high.
- Above USD 3,000: Travel Rule data must travel with the transaction.
- Unhosted wallet: generally allowed, reviewed when risk is high.

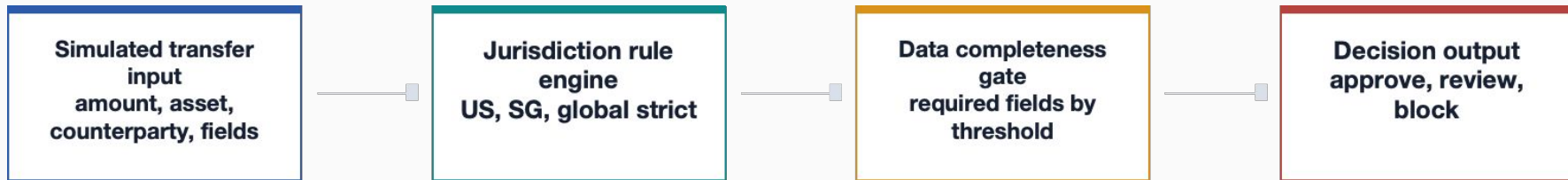
SINGAPORE CONFIGURATION

- Below S\$1,500: basic originator and beneficiary data retained.
- Above S\$1,500: enhanced data required.
- Unhosted wallet: enhanced due diligence is the default posture.



Architecture: rules are separate from risk scoring

The tool knows which law applies before it decides whether a transaction is risky.



Human judgement is required where automation would hide the real normative choice: sanctions hits, high-risk unhosted wallets, disputed blocks, rule changes during an audit period, and cross-border contradictions.

The prototype keeps this boundary explicit by returning a reason code for every transaction, not just a score.

Same transfer, different operational meaning

This is the core requirement: the output changes when the active jurisdiction changes.

Example: USD 1,500 USDC transfer from a US customer to Singapore, hosted beneficiary, no sanctions hit, complete basic identity data.

Active rule	Threshold result	Tool output	Senior-management meaning
US	Below USD 3,000	APPROVE	Velocity is preserved unless AML risk independently escalates.
Singapore	Above S\$1,500	APPROVE WITH ENHANCED DATA	Extra evidence must be collected/transmitted; friction is a feature.
Global strict	Use strictest applicable rule	APPROVE WITH ENHANCED DATA	Avoid regulatory arbitrage in cross-border flows.

Design Choice

The prototype defaults to global strict mode when a transfer touches multiple jurisdictions, but it shows the business cost of that choice instead of pretending stricter always means better.

Synthetic prototype result: Singapore creates materially more work

500 synthetic transfers generated to stress the USD 1,110 to USD 3,000 conflict band.



Enhanced means the transaction crosses the active jurisdiction's Travel Rule threshold and requires the higher information set.

90.4%

transactions where US and SG outputs differ in some way

41.2%

SG false-positive proxy rate in synthetic labels

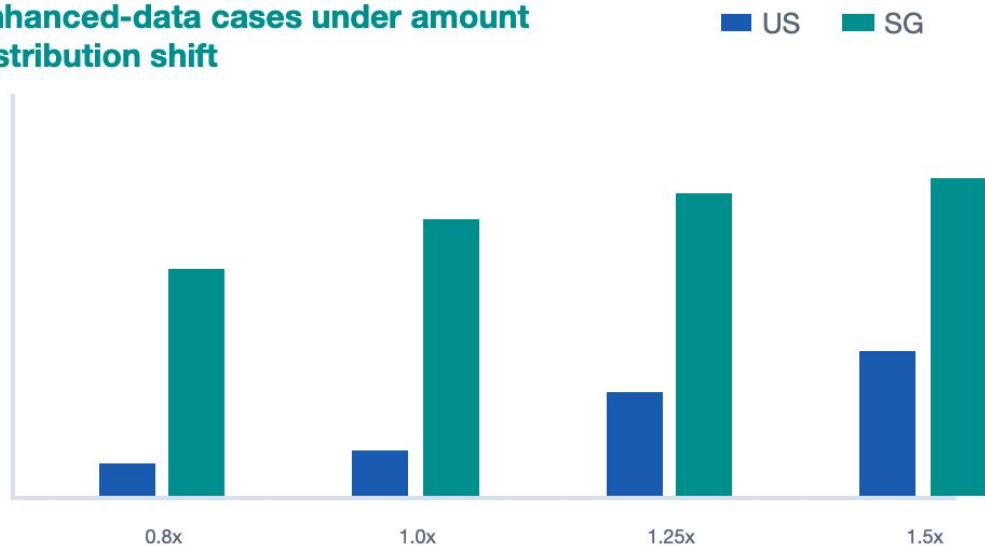
The honest management story: stricter Singapore logic reduces simulated false negatives but pushes many more legitimate users into review or block. That consumer friction is not a bug; it is the cost of the chosen compliance posture.



Sensitivity: conclusions move when transaction sizes shift

The threshold design is exposed to market conditions, not just legal interpretation.

Enhanced-data cases under amount distribution shift



If crypto transfer sizes rise, the US catches up because more transfers cross USD 3,000. But Singapore remains stricter at every simulated shift.

This is why the product must version thresholds and rerun prior-period impact analysis after rule or market changes.



Failure modes are management decisions, not edge cases

The system should surface uncertainty instead of hiding it behind a compliance score.

RULE CHANGES MID-PERIOD

Re-evaluate affected cases under old and new rule versions.

MODEL DRIFT

Monitor review overturn rates and regional false-positive proxies.

CONTRADICTIONARY JURISDICTIONS

Default to strict mode; escalate when consumer harm is material.

MISSING CUSTOMER DATA

Block only when legally required; otherwise queue remediation.

Boundary choice: this prototype does not claim to identify criminals. It routes transactions and evidence obligations. Suspicion remains a human and institutional judgement.



Governance stub: what must be documented before launch

The model card is deliberately plain because the risk is operational misunderstanding.

INTENDED USE

Route Coinbase-style DPT transfers under US, SG and global strict Travel Rule configurations.

INPUTS

Amount, jurisdiction, counterparty type, required-field presence, chain risk, sanctions/adverse media flags.

NOT INTENDED USE

Standalone AML detection, sanctions screening, consumer explanation or legal advice.

MONITORING

Alert quality, reviewer overturns, confirmed suspicious activity, consumer harm, data completeness, regional bias.

Version control is the compliance feature: every decision should be reproducible against the exact rule configuration active at the time.



Recommendation: build the rule engine, but sell it honestly

Greenlight a narrow prototype; do not market it as an AML oracle.

GO

Proceed with a rules-first Travel Rule control layer for Coinbase-style cross-border DPT flows. The product is right for NEXUS because it turns regulatory fragmentation into repeatable infrastructure.

Initial scope: US, Singapore, global strict mode, synthetic-data demo, model card and audit trail.

DO NOT CLAIM

Do not claim the prototype detects laundering, solves fairness, or eliminates human judgement. Its value is jurisdiction routing and evidence discipline.

With more time: live legal update workflow, case-management integration, bias dashboard, reviewer QA and regulator-specific export packs.